BuckleMap: Rotational Restraint, Eigenvalue Error, and Sensitivity in an Elastic Column

Reproducible computational study, October 2026

Abstract

Rotational end restraint can substantially increase an ideal elastic buckling load, but a well-solved discrete eigenproblem alone does not establish an accurate continuum prediction. This study implements exactly integrated cubic Hermite elements in C++17, with Python assembly, analytical verification and visualization. An independent scalar continuum equation tests equal end springs, while four classical support conditions test constraint handling. On nested meshes, the last measured load-convergence orders are 3.9979 for pinned ends and 3.9904 for dimensionless spring stiffness ten. At four elements, the latter case has 0.2698% load error despite a normalized matrix residual of 6.123×10^{-17}. Modal sensitivity agrees with the continuum derivative and exhibits diminishing restraint benefit. These results verify an ideal numerical model; they neither establish physical validity nor provide safe design loads.

1 Question, model and assumptions

The question is how end restraint changes the first column eigenload, and how to distinguish discretization error from algebraic solution error. Hermite interpolation and linearized pre-buckling are established finite-element concepts [1, 2]; no new buckling theory is proposed here. Consider a straight planar column of length L, positive rigidity EI, and a uniform conservative compressive dead load P. The small bifurcation displacement w follows Euler–Bernoulli kinematics, $\theta = w'$; shear deformation, material nonlinearity, imperfections and postbuckling are excluded. Rotational end springs have stiffnesses $k_0, k_L \geq 0$. The quadratic potential is

$$\Pi_2(w) = \tfrac{1}{2}\int_0^L EI(w'')^2\,dx + \tfrac{1}{2}\sum_{s\in\{0,L\}} k_s w'(s)^2 - \tfrac{P}{2}\int_0^L (w')^2\,dx. \tag{1}$$

Its stationary weak form, for admissible test functions v, is

$$a(w,v) = Pg(w,v), \quad a(w,v) = \int_0^L EIw''v''\,dx + \sum_s k_s w'(s)v'(s), \quad g(w,v) = \int_0^L w'v'\,dx. \tag{2}$$

The strong equation within each constant-rigidity segment is $EIw'''' + Pw'' = 0$. The units of EI are force times length squared; those of k_s are force times length. Spring energy enters the elastic form, not the destabilizing geometric form.

2 Independent continuum reference

For uniform rigidity, $w(0) = w(L) = 0$ and equal springs, set $\kappa = k_sL/EI$ and $z = \sqrt{PL^2/EI}$. The symmetric first mode and its end conditions give

$$w(x) = A\{\cos[z(x/L - 1/2)] - \cos(z/2)\}, \tag{3}$$

$$\kappa\sin(z/2) + z\cos(z/2) = 0, \qquad \pi \leq z \leq 2\pi. \tag{4}$$

In particular, $EIw''(0) = k_sw'(0)$ and $EIw''(L) = -k_sw'(L)$. Brent's scalar root method yields $P = EIz^2/L^2$ independently of the FEM matrices. The endpoints recover pinned π^2EI/L^2 and clamped $4\pi^2EI/L^2$ loads. This derivation uses the classical column energy model [3]; it is a reference under the same physical assumptions, not an independent physical experiment.

3 Discretization and implementation

Each two-node element has DOFs $q_e = (w_1, \theta_1, w_2, \theta_2)^T$ and cubic Hermite interpolation $w_h = Nq_e$. Exact integration gives

$$K_e = \int EI(N'')^T N''\,dx, \qquad G_e = \int (N')^T N'\,dx, \qquad (K - PG)q = 0. \tag{5}$$

A dependency-free C++17 kernel supplies the element matrices through pybind11. Python validates input, assembles both matrices, adds spring stiffnesses to rotation diagonals, and eliminates constrained DOFs. The congruence scaling $D_{ii} = K_{r,ii}^{-1/2}$ reduces mixed displacement/slope conditioning. SciPy's symmetric definite eigensolver [4] solves $(DK_rD)y = P(DG_rD)y$ and recovers $q_r = Dy$. The modes are G-orthonormal. No penalty parameter is needed. Nonuniform meshes and elementwise constant rigidity are supported, but the accuracy study uses uniform values.

For a simple first eigenvalue, differentiation with respect to equal springs gives, at $L = EI = 1$,

$$\frac{dP_h}{d\kappa} = q^T \frac{dK}{d\kappa} q = \theta_0^2 + \theta_L^2, \qquad q^T G q = 1. \tag{6}$$

The continuum derivative follows by implicit differentiation of Eq. (4), independently of the discrete eigenvectors.

4 Verification setup and measured results

All experiments use $L = EI = 1$ and double precision. Convergence sequences use $n = 4, 8, 16, 32$ elements at $\kappa = 0$ and 10. The restraint sweep uses 32 elements and $\kappa = 0$ plus 25 logarithmically spaced values from 10^{-2} to 10^4. Table 1 checks four support presets. The clamped–pinned reference is z^2, with $\tan z = z$, $z \in (4.4, 4.6)$. The cantilever is fixed at its left end. These cases exercise different essential constraints and natural boundary conditions.

Table 1: First loads on 32 elements. Errors concern the ideal continuum.

Support	Continuum	FEM	Relative error
pinned	9.86960440	9.86960567	1.290×10^{-7}
clamped	39.47841760	39.47849894	2.060×10^{-6}
cantilever	2.46740110	2.46740112	8.146×10^{-9}
clamped-pinned	20.19072856	20.19073945	5.393×10^{-7}

Field error is the $L^2(0,1)$ difference after separately scaling analytical and FEM symmetric modes to unit midpoint displacement; eight Gauss points per element integrate its square. Algebraic error is measured on the original reduced matrices:

$$\eta = \frac{\|K_r q_r - P_h G_r q_r\|_\infty}{(\|K_r\|_\infty + P_h\|G_r\|_\infty)\|q_r\|_\infty}. \tag{7}$$

This normwise, scaling-dependent backward indicator is not an eigenvalue forward-error bound. Table 2 separates it from continuum load and reconstructed-field error.

Table 2: Mesh errors: $e_P = P_h/P - 1$, unit-peak field error e_w, and algebraic indicator η.

κ	n	e_P	e_w	η
0	4	5.121×10^{-4}	4.430×10^{-4}	6.503×10^{-17}
0	8	3.277×10^{-5}	2.785×10^{-5}	6.113×10^{-17}
0	16	2.060×10^{-6}	1.744×10^{-6}	2.521×10^{-17}
0	32	1.290×10^{-7}	1.090×10^{-7}	4.553×10^{-17}
10	4	2.698×10^{-3}	1.725×10^{-3}	6.123×10^{-17}
10	8	1.895×10^{-4}	1.107×10^{-4}	9.413×10^{-17}
10	16	1.217×10^{-5}	6.977×10^{-6}	5.794×10^{-17}
10	32	7.655×10^{-7}	4.370×10^{-7}	8.086×10^{-17}

5 Restraint benefit and numerical interpretation

Figure 1 shows load and field convergence alongside the matrix residual. Final load orders $\log_2(e_P(16)/e_P(32))$ are 3.9979 and 3.9904. At $\kappa = 10$, the 32-element load is 28.16771809; the continuum gain relative to pinned ends is 2.85398. The four-element spring case still has 0.2698% load error at $\eta = 6.123 \times 10^{-17}$. Increasing eigensolver precision would not remove this discretization error.

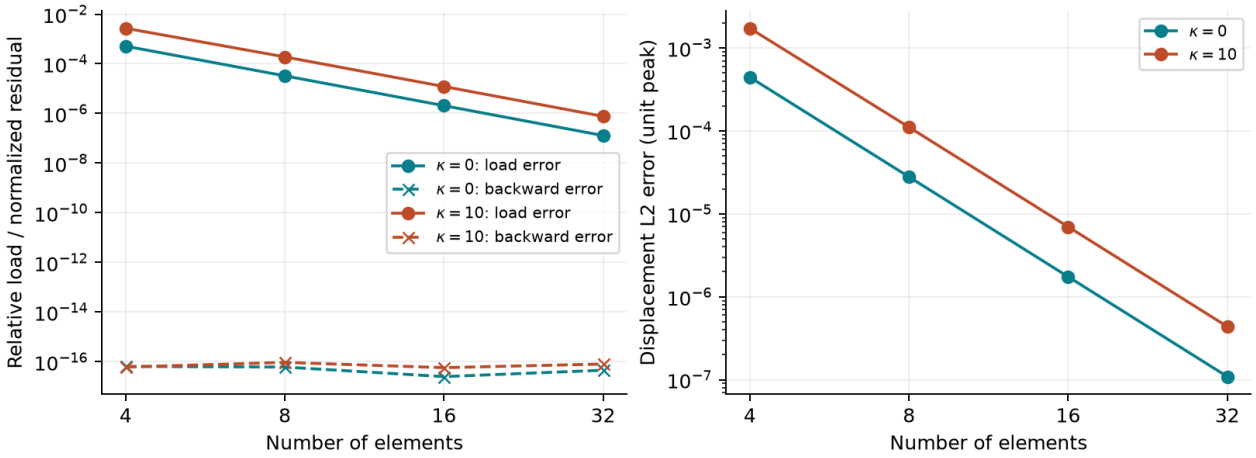


Figure 1: Load versus algebraic error (left) and Hermite field error (right). Curves are measured outputs; residuals depend on platform rounding.

For conforming Hermite spaces and exact integration, the min–max principle gives Ritz eigenvalues above ideal continuum values in exact arithmetic. Nested refinement therefore decreases the first load toward the continuum answer. The observed sequences are consistent with this property; the floating-point implementation does not certify rigorous bounds. Field reconstruction provides an additional check beyond a scalar eigenload.

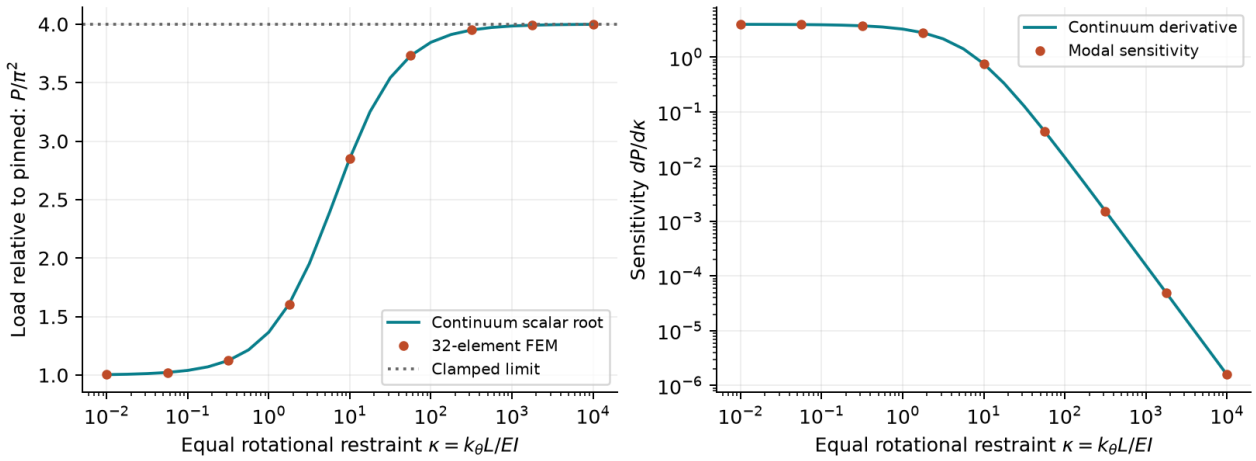


Figure 2: Equal spring restraint raises the first load toward the clamped limit, while sensitivity decreases. Curves use the continuum root and its derivative; markers use the 32-element eigenproblem.

The sensitivity agreement in Fig. 2 supports the sign and placement of spring contributions. Saturation follows Eq. (4); it is not evidence that a real connection has a linear, load-independent rotational stiffness. Equation (6) assumes a simple eigenvalue; repeated eigenvalues require analysis of their invariant subspace.

6 Reproducibility and software checks

The repository contains a native CMake/CTest target, an installable Python package, CLI, and manuscript sources. The command `bucklemap study -output output/study` regenerates the synthetic measurements and figures. CSV and JSON retain unrounded values. The manuscript builder derives every result table and numerical macro from the JSON. No private data or course handouts are required. Topic-level course inspiration is documented in the README; the code, figures and prose were authored for this investigation.

The Python suite checks native matrices against independently fitted physical polynomials and Gauss integration, cubic reconstruction, Euler references, higher modes and geometric orthogonality, dimensional scaling, heterogeneous nested refinement, spring monotonicity, modal derivatives against central finite differences, invalid inputs and CLI behavior. Native checks exercise rigid modes and symmetry. C++ warnings are errors; formatting checks cover both languages. CI builds wheel and source distribution, tests the installed wheel on Windows and Linux, and reproduces the numerical records. These are checks of specified cases, not an exhaustive proof over permitted inputs.

The local environment uses Python 3.14 and MSVC 19.44; package versions are in `requirements-tested.txt`. Cross-platform reproduction tolerances are 3×10^{-8} relative and 3×10^{-10} absolute for ordinary numeric fields, a 10^{-12} ceiling for η, and 0.02 for derived convergence orders. These allowances reflect BLAS-dependent rounding; they do not redefine the continuum errors reported in Table 2.

The command `python paper/build.py -engine pdflatex` regenerates result macros, compiles twice, rejects undefined references and overfull boxes, and enforces a five-page maximum including references. Tectonic is also supported. A SHA-256 manifest binds the PDF to source, solver files, data and figures. CI renders all pages for inspection. Visual review is needed in addition to automatic log and page-count checks. Identical PDF bytes across engines are not required.

7 Limitations and conclusion

Dense storage is quadratic and the eigensolve cubic in the free-DOF count, so the implementation targets small studies. Diagonal scaling reduces mixed-unit conditioning problems but cannot guarantee accuracy for extreme mesh ratios, restraint stiffnesses or rigidity contrasts. Only a uniform conservative axial force is supported. Follower forces, shear flexibility, initial crookedness, plasticity, nonlinear joints and load redistribution are excluded. Agreement with the Euler model does not calibrate physical behavior; real columns can fail below these ideal eigenloads. They must not be interpreted as allowable loads or design-code compliance.

Within those assumptions, the experiments show diminishing benefit from rotational restraint, nearly fourth-order load convergence on the tested meshes, and agreement of modal and continuum sensitivities. They also show why algebraic residual, discretization error and physical validity must be assessed separately. BuckleMap is a reproducible implementation and verification study of established mechanics, without claims of theoretical novelty, affiliation or peer review.

References

[1] TU Delft. *Euler–Bernoulli beam elements*. Finite Elements in Civil Engineering and Geosciences, Sec. 4.1.

[2] TU Delft. *Geometrically nonlinear problems*. Finite Elements in Civil Engineering and Geosciences, Sec. 5.4.

[3] TU Delft OpenCourseWare. *Stability of Beams*. Buckling of Structures, lecture resource.

[4] SciPy developers. *scipy.linalg.eigh*. SciPy API reference.